Assignment 8

Remote Group

For November 8th 2026

Problem (Munkres 52.6)**.** Show that if X is path connected, the homomorphism induced by a continuous map is independent of base point, up to isomorphism of the groups involved. More precisely, let $h : X \to Y$ be continuous, with $h(x_0) = y_0$ and $h(x_1) = y_1$. Let α be a path in X from x_0 to x_1, and let $\beta = h \circ \alpha$. Show that

$$\hat{\beta} \circ (h_{x_0})_* = (h_{x_1})_* \circ \hat{\alpha}.$$

This equation expresses the fact the following diagram of maps commutes.

$$\begin{array}{ccc} \pi_1(X, x_0) & \xrightarrow{(h_{x_0})_*} & \pi_1(Y, y_0) \\ \downarrow{\scriptstyle \hat{\alpha}} & & \downarrow{\scriptstyle \hat{\beta}} \\ \pi_1(X, x_1) & \xrightarrow{(h_{x_1})_*} & \pi_1(Y, y_1) \end{array}$$

Proof. As X is path connected, there exists $\alpha : [0, 1] \to X$ with $\alpha(0) = x_0,\ \alpha(1) = x_1$. α induces a change-of-base-point isomorphism

$$\hat{\alpha} : \pi_1(X, x_0) \to \pi_1(X, x_1),$$

by

$$[f] \mapsto [\bar{\alpha} * f * \alpha].$$

Similarly, h induces

$$(h_{x_0})_*([f]) = [h \circ f].$$

Let $[f] \in \pi_1(X, x_0)$. Applying $(h_{x_0})_*$ gives, as mentioned,

$$[f] \mapsto [h \circ f].$$

Applying $\hat{\beta}$ gives

$$\hat{\beta}((h_{x_0})_*([f])) = [\bar{\beta} * (h \circ f) * \beta].$$

Applying $\hat{\alpha}$ gives, as mentioned,

$$[f] \mapsto [\bar{\alpha} * f * \alpha].$$

Applying $(h_{x_1})_*$ gives

$$(h_{x_1})_*(\hat{\alpha}([f])) = [h \circ (\bar{\alpha} * f * \alpha)].$$

Now it remains to show that these expressions are equal.

Since $\beta = h \circ \alpha$ we have that

$$h \circ \bar{\alpha} = \overline{h \circ \alpha} = \bar{\beta}.$$

For continuous maps,

$$h \circ (\gamma_1 * \gamma_2) = (h \circ \gamma_1) * (h \circ \gamma_2).$$

Therefore

$$\begin{aligned} h \circ (\bar{\alpha} * f * \alpha) &= (h \circ \bar{\alpha}) * (h \circ f) * (h \circ \alpha) \\ &= \bar{\beta} * (h \circ f) * \beta. \end{aligned}$$